

Predictive Value of Biological Sex for Determining Level of Care in Symptomatic Poisoned Patients

Yash Jaggi

Key Results for Abstract

1. Demographics

Original external validation set (n=105): 56 females (53%), median age 28 [IQR 18–44]; 49 males (47%), median age 31 [IQR 22–50]

Larger data set (n=189): 101 females (53%), median age 29 [IQR 20–47]; 88 males (47%), median age 32 [IQR 26–51]

2. Sex-Exposure Association (N= 219)

Overall chi-squared test: $p = 0.007$

- Antidepressants: 78% female, 22% male
- Street Drugs: 35% female, 65% male
- CO, As, CN: 39% female, 61% male

3. Model Performance (INTOXICATE score > 6)

Deriving the male adjustment: glm(ICU ~ INTOXICATE_SCORE + Gender) on the larger dataset yields a sex coefficient of 1.29 [0.45–2.14], $p=0.003$. Scaling to the INTOXICATE points system gives +3 points for biological males.

Original external validation set (n=105):

	Sensitivity	95% CI	Specificity	95% CI
Without sex	0.667	0.454–0.828	0.429	0.328–0.535
With sex (+3 male)	0.905	0.711–0.973	0.357	0.263–0.464

Larger data set (n=189):

	Sensitivity	95% CI	Specificity	95% CI
Without sex	0.737	0.580–0.850	0.576	0.496–0.652
With sex (+3 male)	0.895	0.759–0.958	0.457	0.380–0.536

4. Regression Coefficients (for reference)

Original (n=105): Sex coefficient 1.895 [0.752–3.260], $p=0.002$

Combined (n=189): Sex coefficient 1.292 [0.471–2.178], $p=0.003$

Reference — INTOXICATE-US paper (adults only, n=77, score > 6): Sensitivity: 0.69 [0.41–0.89], Specificity: 0.38 [0.26–0.51]

Sex-Exposure Table

Table 3: Sex vs type of xenobiotic (N = 219, chi-squared p = 0.007).

	Biological Sex		Total	p-value	Row vs Rest p	Row vs Rest p (adj)
	<i>F</i>	<i>M</i>				
Exposure Category						
<i>Combination</i>	23 (40%)	34 (60%)	57 (100%)		0.097	0.258
<i>Analgesic</i>	17 (57%)	13 (43%)	30 (100%)		0.611	0.698
<i>CO, As, CN</i>	11 (39%)	17 (61%)	28 (100%)		0.276	0.441
<i>Unknown</i>	15 (54%)	13 (46%)	28 (100%)		0.901	0.901
<i>Antidepressants</i>	21 (78%)	6 (22%)	27 (100%)		0.005	0.041
<i>Street Drugs</i>	7 (35%)	13 (65%)	20 (100%)		0.216	0.432
<i>Alcohol</i>	6 (40%)	9 (60%)	15 (100%)		0.555	0.698
<i>Sedatives</i>	11 (79%)	3 (21%)	14 (100%)		0.060	0.240
Total	111 (51%)	108 (49%)	219 (100%)			